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Named Entity-Driven Graph Smoothing to Enhance Pretrained Document Embeddings in Clustering Tasks

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Abstract

Named Entity Recognition (NER) extracts structured semantic elements, such as persons, organizations, or biomedical concepts, that enhance document understanding. Rather than replacing pretrained document embeddings, we refine them by injecting entity-level relational information. By constructing a document graph based on named-entity similarity, we apply spectral filtering to smooth the embedding matrix, yielding entity-aware representations compatible with any clustering algorithm. This framework is model-agnostic and lightweight. Furthermore, it can distill a set of high-confidence Must-Link (ML) constraints, pairs of documents that should be assigned to the same cluster, to guide constrained clustering. Experimental results on benchmark datasets demonstrate that our approach significantly improves clustering performance, highlighting the value of entity-aware smoothing over standard pretrained representations.

CCS Concepts

• **Information systems** → **Clustering and classification**; • **Computing methodologies** → *Information extraction*; *Semi-supervised learning settings*.

Keywords

Information Retrieval, Clustering, Unsupervised Learning, Semi-supervised Learning, Large Language Models, Named Entity Recognition

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1 Introduction

Document clustering lies at the core of many information retrieval and analytics systems [1, 2, 5, 16, 27, 28]. Despite its importance, standard pipelines often rely heavily on surface-level lexical signals such as TF-IDF [1, 3] or dense embeddings paired with simple distance metrics [4, 29]. Although Large Language Models

(LLMs) [26, 31] have significantly improved document representations, their use in unsupervised learning remains challenging. In heterogeneous corpora, topical boundaries can blur when distinctions hinge more on *who/what/where* than on shared surface phrasing, particularly when texts are short, multilingual, or stylistically diverse [6, 7, 14, 20, 25].

We argue that structured semantic signals, in particular, named entities [21], offer stable topical anchors that complement contextual embeddings. Entities refer to persistent real-world objects (e.g., persons, organizations, statutes, products, or biomedical concepts), and even a single mention can strongly cue domain membership (e.g., “Pfizer”, “Cristiano Ronaldo”, or the “State Council”). As illustrated in Figure 1, entity evidence can link thematically related documents that share little vocabulary. At the same time, entities alone rarely suffice to recover fine-grained thematic structure; discarding the surrounding lexical/contextual signals typically hurts performance. This motivates entity-aware clustering that treats entities as complementary structures rather than replacements for text.

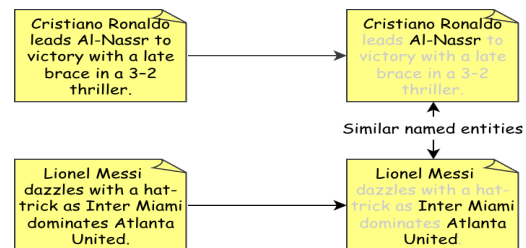


Figure 1: Illustration of the importance of named entities.

We propose a minimal, type-aware NER graph that captures entity similarity between documents and integrates it into off-the-shelf clustering. Specifically, we introduce a unified pipeline that: (a) smooths LLM-based features with a normalized entity graph ($\tilde{M}_X$) and (b) optionally distills a small set of Must-Link (ML) constraints for constrained clustering. Our empirical study across multiple benchmarks shows consistent gains over embedding-only baselines, with the most significant improvements observed on entity-dense corpora.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 details the entity-aware graph construction and constraint distillation; Section 4 provides a complexity analysis of the proposed framework; Section 5 presents the experimental setup; Section 6 analyzes the results and visualizations; and Section 7 concludes.



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2 Related Work

Despite advances in document embeddings and contextual language models, named entities remain underused in document clustering. Nevertheless, prior work demonstrates their capacity to act as stable semantic anchors in lexically diverse corpora, primarily by enriching vector space models with features such as entity names, types, and knowledge base identifiers. Cao et al. [12] adapted the vector space model to integrate these entities as structured features, showing improved performance on topic-specific corpora. Their experiments indicate that documents sharing semantically similar entities tend to be more thematically aligned, even when surface vocabulary diverges. In a complementary direction, Alsudais and Tchalian [8] explored the clustering of entities themselves, particularly persons and organizations, to reveal hidden relationships within a topical corpus. While their approach focuses on entity co-occurrence patterns, it implicitly supports document-level clustering by uncovering latent thematic structures centered on prominent actors. Other research has examined how the presence of such structured signals affects the interpretability of topic models. Tolochko et al. [30] found that preserving named entities improves the coherence and semantic clarity of discovered topics, highlighting their importance as discriminative signals in textual data.

Constrained clustering integrates prior knowledge to guide the optimization process, enhancing both cluster quality and interpretability. Constraints are typically categorized as either instance-level or cluster-level. In this work, we focus on instance-level signals, specifically Must-Link (ML) and Cannot-Link (CL) constraints. While an ML constraint mandates that two documents occupy the same cluster, a CL constraint ensures they remain separate. We employ two foundational formulations: COP K-means [33], which enforces hard feasibility, and PC K-means [10], which incorporates soft penalties. To better handle high-dimensional text embeddings, we also implement spherical variants by substituting Euclidean distance with cosine similarity.

COP K-means. We minimize within-cluster inertia under global feasibility constraints:

$$\min_{\mathbf{z}, \{\mu_g\}} \sum_{i=1}^N \sum_{g=1}^G z_{ig} \|\mathbf{x}_i - \mu_g\|^2. \quad (1)$$

Here, $z_{ig} \in \{0, 1\}$ is a binary assignment indicator where $\sum_g z_{ig} = 1$. The values of z_{ig} are determined during the assignment phase: for each document $\mathbf{x}_i$, the algorithm sets $z_{ig} = 1$ for the nearest cluster g that satisfies all ML and CL constraints. Specifically, if $(\mathbf{x}_i, \mathbf{x}_j) \in \text{ML}$ and $\mathbf{x}_j$ is already assigned to cluster g' , then $\mathbf{x}_i$ must be assigned to g' ($z_{ig'} = 1$), even if $\mathbf{x}_i$ is geometrically closer to another centroid.

PC K-means. In the soft-constrained version, z_{ig} is calculated by minimizing the sum of the squared Euclidean distance and the cost of constraint violations. The objective function integrates penalties for ML and CL violations with weights $\lambda_{\text{ML}}, \lambda_{\text{CL}} \geq 0$:

$$\begin{aligned} \min_{\mathbf{z}, \{\mu_g\}} & \sum_{i=1}^N \sum_{g=1}^G z_{ig} \|\mathbf{x}_i - \mu_g\|^2 + \lambda_{\text{ML}} \sum_{(i,j) \in \text{ML}} \left(w_{ij}^{\text{ML}} \left[1 - \sum_{g=1}^G z_{ig} z_{jg} \right] \right) \\ & + \lambda_{\text{CL}} \sum_{(i,j) \in \text{CL}} \left(w_{ij}^{\text{CL}} \sum_{g=1}^G z_{ig} z_{jg} \right). \end{aligned} \quad (2)$$

where w_{ij}^{ML} and w_{ij}^{CL} denote the confidence levels associated with each pairwise constraint. In this case, the assignment $z_{ig} = 1$ results from a trade-off: a document might be assigned to a slightly more distant cluster if doing so prevents a high-penalty constraint violation.

3 Proposed Method: NER-Based Graph

Our preliminary experiments suggest that named entities should serve as complementary signals for clustering regularization rather than replacing the underlying text. To this end, we propose a simple and effective graph-based framework that combines named entities with semantic embeddings for document clustering (Figure 2).

In our approach, the node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, traditionally representing n documents described by d features, is replaced by LLM embeddings, modifying d to d_{ellm} .

Let $f_{\text{ellm}} : \mathcal{D} \rightarrow \mathbb{R}^{d_{\text{ellm}}}$ be the LLM model that maps each document $d_i \in \mathcal{D}$ or its components (words, sentences, or entities) to a vector in a high-dimensional embedding space. The feature matrix $\mathbf{X}_{\text{ellm}}$ is constructed as:

$$\mathbf{X}_{\text{ellm}} = \begin{bmatrix} g_{\text{ellm}}(d_1) \\ g_{\text{ellm}}(d_2) \\ \vdots \\ g_{\text{ellm}}(d_n) \end{bmatrix} = [\mathbf{x}'_1, \dots, \mathbf{x}'_{d_{\text{ellm}}}] \in \mathbb{R}^{n \times d_{\text{ellm}}} \quad (3)$$

where each row $g_{\text{ellm}}(d_i) = \mathbf{x}_i$ is the embedding of document d_i .

Unlike traditional clustering methods based only on text similarity, our model leverages named entities as semantic anchors to improve document connectivity and refine similarity computation.

Named Entity Recognition (NER). Given a dataset $\mathcal{D}$ containing n documents $\mathcal{D} = \{d_1, \dots, d_n\}$, we apply a NER model to extract entities from each document d_i :

$$\mathcal{E}_{d_i} = \{e_1, \dots, e_{|\mathcal{E}_{d_i}|}\}, \quad \forall d_i \in \mathcal{D} \quad (4)$$

Entity Similarity Computation. To establish meaningful relationships between documents, we compute entity-based document similarity. Each named entity $e \in \mathcal{E}$ is mapped to a vector representation using a trained word embedding model such as Word2Vec or fast-Text [11]. The entity embedding function is defined as $g : \mathcal{E} \rightarrow \mathbb{R}^d$ where $f(e)$ is the embedding vector of entity e . The similarity between two documents d_i and d_j is computed as the sum of cosine similarities of their named entities, considering different entity types:

$$\text{Sim}(d_i, d_j) = \sum_{t \in T} \sum_{e \in E_t} \cos(f(e_i), f(e_j)) \quad (5)$$

where T represents the set of entity types, E_t is the set of named entities of type t , and $\cos(f(e_i), f(e_j))$ is the cosine similarity between the corresponding entities. This ensures that documents sharing multiple strong entity connections receive higher scores.

Graph Construction and Normalization. Using the similarity matrix computed from entities, we construct the document graph, where each node corresponds to a document, and an edge is created between two nodes based on their similarity scores. The adjacency matrix $\mathbf{M} = (m_{ij}) \in \mathbb{R}^{n \times n}$ is defined by $m_{ij} = \text{Sim}(d_i, d_j)$ where $\text{Sim}(d_i, d_j)$ is the sum of cosine similarities between entities of the

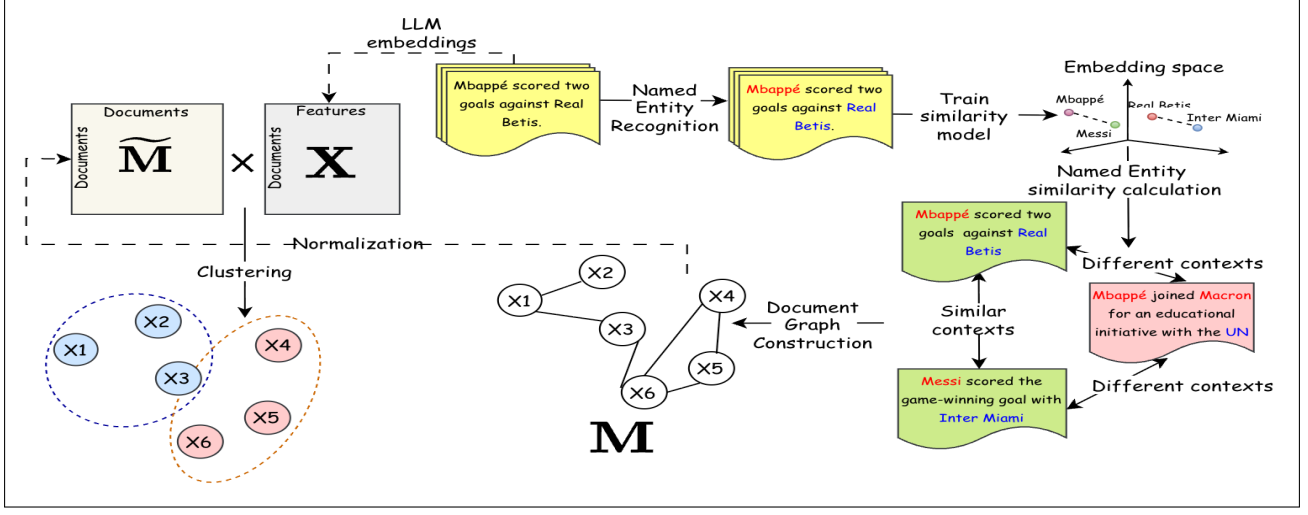


Figure 2: Overview of the proposed method: LLM-based feature extraction, NER-based graph construction, and clustering.

same type in d_i and d_j , counting only pairs with cosine similarity $\geq \tau$ (0.7 in our experiments).

Graph Smoothing and Constraint Extraction. We construct the normalized adjacency matrix:

$$\tilde{\mathbf{M}} = \mathbf{D}^{-1/2} (\mathbf{M} + \beta \mathbf{I}) \mathbf{D}^{-1/2}, \quad (6)$$

where $D_{ii} = \sum_j m_{ij}$ and $\beta > 0$, and apply a propagation over document embeddings to obtain smoothed representations $\mathbf{X} \leftarrow \tilde{\mathbf{M}}\mathbf{X}$.

In addition, we exploit the entity similarity matrix $\mathbf{M}$ for constraint extraction. We interpret it as a weighted similarity graph and define the set of candidate pairs as $\mathcal{P} = \{(i, j) : 1 \leq i < j \leq n, m_{ij} > 0\}$. We then select the top- b strongest edges among these pairs to form ML constraints, denoted $\text{ML} = \text{Top-}b(m_{ij} \mid (i, j) \in \mathcal{P})$, with confidence weights $w_{ij} = m_{ij}$.

4 Complexity Analysis

The computational cost is mainly driven by three stages: NER and embedding, graph construction, and feature smoothing, assuming a highly sparse similarity matrix $\mathbf{M}$, as only a few documents typically share many named entities.

Theoretical Complexity. In a theoretical worst-case scenario where $\mathbf{M}$ is dense:

- **Graph Construction:** Computing the similarity between n documents requires $O(n^2 \cdot |\mathcal{E}_{\max}|)$ operations, where $|\mathcal{E}_{\max}|$ is the maximum number of entities per document.
- **Smoothing:** The matrix multiplication $\tilde{\mathbf{M}}\mathbf{X}$ involves a product of $(n \times n)$ and $(n \times d_{\text{ellm}})$ matrices, leading to a complexity of $O(n^2 d_{\text{ellm}})$.

Practical Complexity. Given that $\mathbf{M}$ is very sparse, let $\text{nnz}(\mathbf{M})$ be the number of non-zero entries in the adjacency matrix ($\text{nnz}(\mathbf{M}) \ll n^2$).

- **NER & Entity Embedding:** This step is $O(n \cdot L)$, where L is the average document length. Since this can be parallelized at the document level, it is highly efficient in practice.

- **Sparse Graph Construction:** By using an inverted index (mapping entities to the documents containing them), we only compute similarities for pairs of documents (d_i, d_j) that share at least one entity. This reduces the construction cost to $O(n \cdot \bar{e} \cdot \bar{d})$, where $\bar{e}$ is the average number of entities per document and $\bar{d}$ is the average degree in the document graph.
- **Sparse Smoothing:** The operation $\mathbf{X} \leftarrow \tilde{\mathbf{M}}\mathbf{X}$ is implemented using sparse-dense matrix multiplication. Its complexity is $O(\text{nnz}(\mathbf{M}) \cdot d_{\text{ellm}})$. Since $\text{nnz}(\mathbf{M}) \approx n \cdot \bar{d}$, the practical complexity is near-linear with respect to the number of documents: $O(n \cdot \bar{d} \cdot d_{\text{ellm}})$.

As d_{ellm} (the embedding dimension) and $\bar{d}$ (average document connectivity) are relatively small constants compared to n , the proposed method scales gracefully to large-scale datasets.

5 Experiments

We assess clustering performance using embeddings and language-specific NER models, enhanced with graph constraints and feature smoothing for robustness.

Setup. We evaluate on five datasets: BBC News ($N=2,225, G=5$), MLSUM ($N=407,835, G=612$), ArXiv-10 ($N=100,000, G=10$), Classic4 ($N=7,095, G=4$), 20 Newsgroups ($N=18,846, G=20$), using Accuracy, NMI, and ARI. Due to computational constraints, we randomly sample 10,000 Arxiv documents and 16,321 MLSUM documents from five categories: Sport, Health, Politics, Economy, and Climate.

Algorithms. We average results over 20 independent runs, using k-means++ [9] initialization, with the number of iterations limited to 300 for K-means [23], SK-means [19], RSDE [22], and Semi-NMF [13]. For deep learning and spectral baselines, namely DK-means [15], IDEC [18], and Spectral Clustering [32], we adhere to the default settings. Finally, to benchmark against recent LLM-driven advancements, we include LLMEdgeRefine [17]. This

method leverages LLMs to iteratively refine boundary points and has been shown to outperform prior approaches like the popular ClusterLLM [35]. To ensure a fair comparison, the true number of clusters is provided to all methods.

Embeddings. We use Qwen3-Embedding-4B [34], generating embeddings of 2560 dimensions. Although larger Qwen3 variants lead current clustering benchmarks, the 4B version offers the best trade-off in our setting.

Named Entity Recognition. Named entities are extracted with models selected to match the language and characteristics of each dataset: for MLSUM (French), we use camembert-ner¹; for BBC News (English), we use DeBERTa; and for the remaining datasets, we rely on GPT-5² with instructions to return entities in JSON format. To represent the extracted named entities as vectors, we choose Word2Vec [24]. This model is selected because of its computational efficiency during training, making it efficient for our use case. Word2Vec is trained using CBOW with num_features = 500, min_word_count = 10, a context window size of 5, and 20 epochs. Parameters are chosen with TensorBoard Projector³ to ensure semantically coherent embeddings.

NER graph: constraints and smoothing. We use the entity-based similarity matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ to derive two components: constraint-based baselines and graph smoothing. We extract $b = 100$ high-confidence ML pairs from $\mathbf{M}$ and ignore CL constraints. For smoothing, we apply a single-step propagation using the normalized adjacency matrix, with $\beta = 1$ in all experiments. These components are evaluated independently and jointly in Table 1.

6 Results Analysis

Before assessing the impact of clustering on smoothed $\tilde{\mathbf{M}}\mathbf{X}$, what about clustering of $\tilde{\mathbf{M}}$? Figure 3 visualizes normalized entity graphs $\tilde{\mathbf{M}}$ for a random subset of 2,000 documents per corpus (nodes = documents, edges = entity-based similarity, colors = ground-truth labels). The graphs display diverse topologies, e.g., compact, well-separated communities in BBC versus denser, more interconnected structures elsewhere, indicating that $\tilde{\mathbf{M}}$ captures meaningful semantic organization and that our smoothing strategy is effective. However, as expected, clustering based on $\tilde{\mathbf{M}}$ alone underperforms the joint use of $\tilde{\mathbf{M}}$ and $\mathbf{X}$ (Table 1), since the graph is intentionally sparse and thus better suited as a high-confidence structural prior that sharpens cluster boundaries rather than as a standalone representation. Quantitatively, the results in Table 1 show that constraint-based variants (COP/PC; spherical forms with cosine dissimilarity) offer a tangible benefit when we distill a few high-confidence must-link constraints from $\mathbf{M}$. On MLSUM, for instance, PC K-means improves from 74.68/58.03/57.42 to 80.20/62.80/63.50 (ACC/NMI/ARI), confirming that constraints do add value, with PC typically more robust than COP. However, at the graph level, directly leveraging entity relations via feature smoothing $\tilde{\mathbf{M}}\mathbf{X}$ yields larger and more consistent gains across algorithms: still in MLSUM, K-means moves from

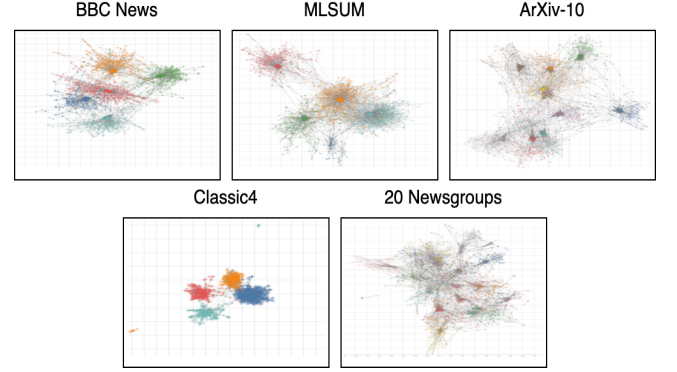


Figure 3: Graph visualizations of $\tilde{\mathbf{M}}$ (2,000-document samples). Nodes are documents, edges encode named-entity similarity, and colors indicate ground-truth classes.

74.68/58.03/57.42 to 85.18/69.59/72.29. $\tilde{\mathbf{M}}\mathbf{X}$ is model-agnostic and exploits the entire graph rather than a small subset of pairs, acting as a light regularizer that pulls semantically close documents together; in our protocol, it is therefore more effective than injecting constraints alone.

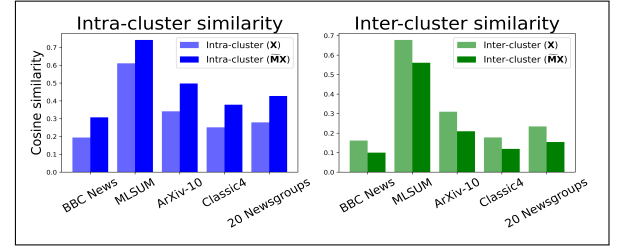


Figure 4: Impact of $\tilde{\mathbf{M}}\mathbf{X}$ vs. $\mathbf{X}$: higher intra-cluster and lower inter-cluster similarity with $\tilde{\mathbf{M}}\mathbf{X}$.

Figure 4 shows that $\tilde{\mathbf{M}}\mathbf{X}$ increases intra-cluster similarity and decreases inter-cluster similarity. The UMAP projections (Figure 5) also show tighter and better separated clusters, corroborating the gains in Table 1 and confirming that the entity-driven graph yields more discriminative embeddings.

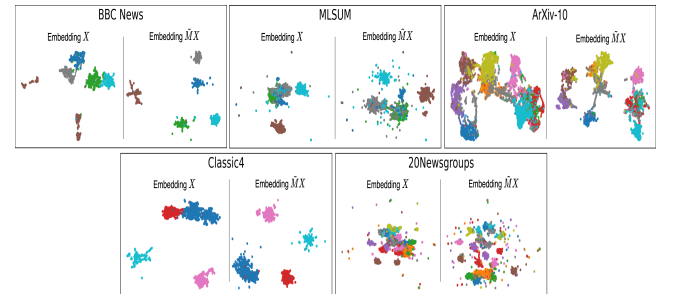


Figure 5: UMAP projections of document representations colored by ground-truth labels: left for $\mathbf{X}$, right for $\tilde{\mathbf{M}}\mathbf{X}$.

¹<https://huggingface.co/Jean-Baptiste/camembert-ner>

²<https://platform.openai.com/docs/models/gpt-5>

³<https://projector.tensorflow.org/>

Table 1: Clustering performance comparison. The values in bold represent the best scores achieved by each algorithm for a given dataset, while the cells highlighted in light violet correspond to the highest scores across all algorithms. Input Type: X (Embeddings only), ML + X (ML constraints applied to embeddings), and $\tilde{M}X$ (Full Entity-aware Graph Smoothing).

Algorithm	Input	BBC			MLSUM			ArXiv-10			Classic4			20 Newsgroups		
		ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
Spectral Clustering	$\tilde{M}$	35.01	20.01	7.87	31.33	1.23	0.20	20.01	6.31	2.12	23.56	19.45	29.34	11.45	9.54	12.45
RSDE	$\tilde{M}$	57.01	42.01	29.87	53.33	23.23	22.20	42.01	28.31	24.12	45.56	41.45	41.34	33.45	31.54	34.45
K-means	X	95.97	89.88	92.72	74.68	58.03	57.42	66.10	61.92	51.74	77.52	72.84	54.71	67.52	69.42	55.00
LLMEdgeRefine	X	95.72	89.35	92.10	73.95	57.10	56.60	65.60	61.10	50.90	76.90	72.10	54.10	66.80	68.90	54.10
COP K-means	ML, X	96.30	90.60	93.30	78.50	60.50	61.00	68.00	63.00	53.00	79.00	74.00	55.80	68.20	70.00	55.90
PC K-means	ML, X	95.70	91.50	93.80	80.20	62.80	63.50	69.20	64.10	54.60	79.80	74.90	56.60	68.90	70.60	56.50
K-means	$\tilde{M}X$	97.01	92.26	93.03	85.18	69.59	72.29	72.59	63.70	62.57	80.83	76.69	56.43	68.67	69.13	58.01
SK-means	X	85.43	82.28	80.02	72.24	56.30	55.19	64.09	59.44	49.10	75.22	72.92	55.62	63.72	68.95	52.77
COP SK-means	ML, X	95.30	90.60	93.30	78.50	60.50	61.00	68.00	63.00	53.00	79.00	74.00	55.80	68.20	70.00	55.90
PC SK-means	ML, X	90.20	84.10	82.90	77.10	60.10	60.20	66.90	61.80	51.70	77.10	73.60	56.10	66.10	69.60	54.30
SK-means	$\tilde{M}X$	92.94	85.61	83.49	84.57	68.50	69.45	68.22	63.03	60.86	78.25	74.50	55.80	64.93	68.28	57.39
DK-means	X	90.01	81.56	82.45	68.67	52.67	48.34	68.24	60.12	52.37	82.11	69.67	60.45	60.35	66.08	49.67
	$\tilde{M}X$	92.42	84.41	84.58	71.39	55.61	50.70	70.83	62.16	55.15	84.77	71.98	62.92	62.54	68.96	51.72
IDEC	X	92.99	82.74	83.74	67.31	50.14	47.05	68.50	60.37	52.06	83.64	69.10	61.53	59.96	65.99	48.96
	$\tilde{M}X$	96.27	84.79	86.29	69.76	53.61	50.40	72.28	62.54	54.90	85.70	71.54	64.54	62.01	68.39	52.26
Semi-NMF	X	88.36	80.07	82.75	82.98	62.28	65.75	75.20	66.27	59.19	79.39	72.52	54.36	60.29	58.01	40.04
	$\tilde{M}X$	95.46	86.97	89.31	84.72	68.75	70.58	72.55	62.64	59.45	80.51	75.23	55.94	63.89	63.08	49.67

7 Conclusion

In this study, we demonstrated that while LLMs provide powerful document representations, they can still struggle with thematic boundaries in heterogeneous corpora. We addressed this by introducing a lightweight, entity-aware framework that leverages structured semantic signals, named entities, as relational anchors. Our findings reveal two key insights. First, the distillation of high-confidence ML constraints from entity-based similarities provides a robust supervisory signal for constrained clustering variants like PC K-means. Second, and more significantly, applying spectral smoothing via the normalized entity graph ($\tilde{M}X$) consistently enhances cluster purity and separation across diverse datasets and algorithms. This model-agnostic approach effectively pulls semantically related documents closer in the embedding space, outperforming standard embedding-only baselines.

This work ultimately demonstrates that named entities serve not just as metadata but as critical structural priors. Building on this, two key research directions emerge: first, integrating Entity Linking to resolve coreferences and enhance graph coherence, and second, applying graph-smoothing techniques to dynamic document streams, where entity stability is crucial. Additionally, further investigation will focus on the granular impact of entity types, investigating whether specific categories (e.g., ORG vs. LOC) provide more discriminative links. We also aim to transition from NER to *Entity Linking* to resolve ambiguities and consolidate synonyms, further densifying the document graph. Finally, leveraging external knowledge bases to compute semantic distances between entities

promises to capture deeper conceptual relationships, potentially leading to even more robust and interpretable document clusters.

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